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GOLD  **1**
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 **2**
GOLD  **1**
IBO
International
Biology Olympiad

 **1**
GOLD  **1**
IOAA
International
Olympiad on
Astronomy &
Astrophysics

 **1**
GOLD  **1**
IOAA (Jr)
International
Olympiad on
Astronomy &
Astrophysics Jr.

 **1**
GOLD  **1**
APhO
Asian Physics
Olympiad



**PROOF THAT
HARD WORK WINS!**

 **2**
SILVER  **2**
IOAA
International
Olympiad on
Astronomy &
Astrophysics

 **1**
SILVER  **2**
APhO
Asian Physics
Olympiad

 **1**
BRONZE  **3**
IOAA (Jr)
International
Olympiad on
Astronomy &
Astrophysics Jr.

 **1**
GOLD  **1**
IJSO
International Junior
Science Olympiad

 **1**
SILVER  **2**
IJSO
International Junior
Science Olympiad

 **2**
SILVER  **2**
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FOCUS



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INSPIRE

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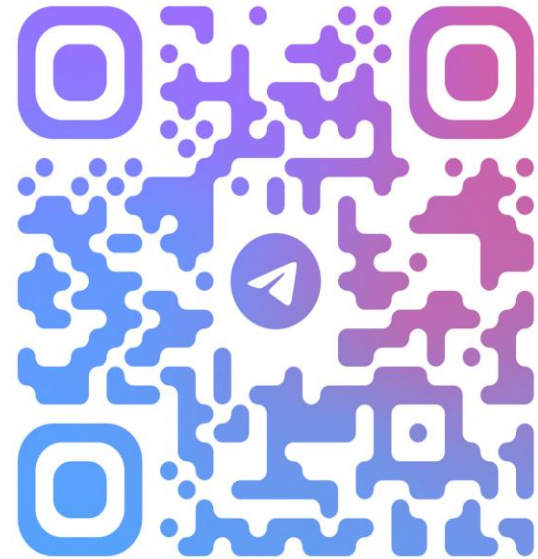


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→ delocalisation
of charge

Types of Resonance or mesomeric effect (R/M)

Notes | Solving content

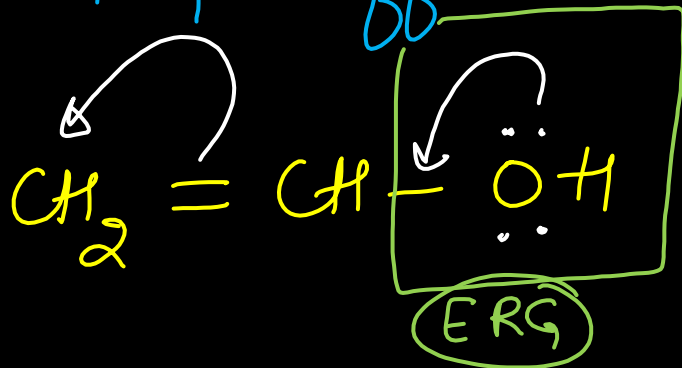
→ Vedantu
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channel

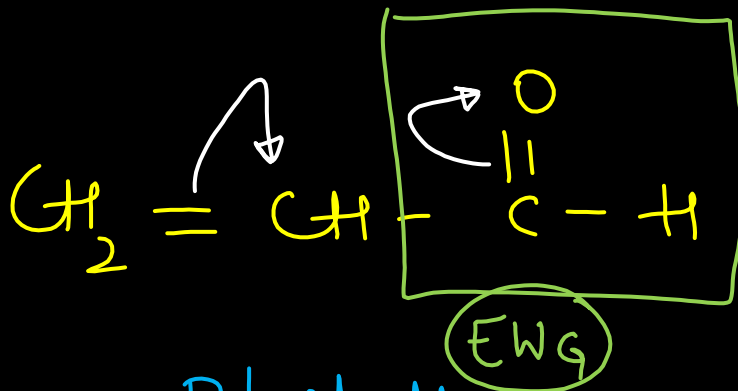
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+R/+M effect

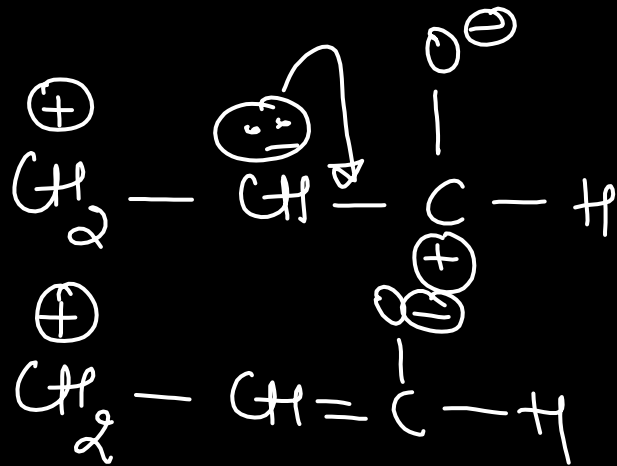
(I)



(II)

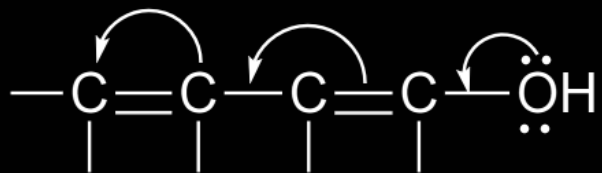


-R/-M effect



Mesomeric effect (R/M) :

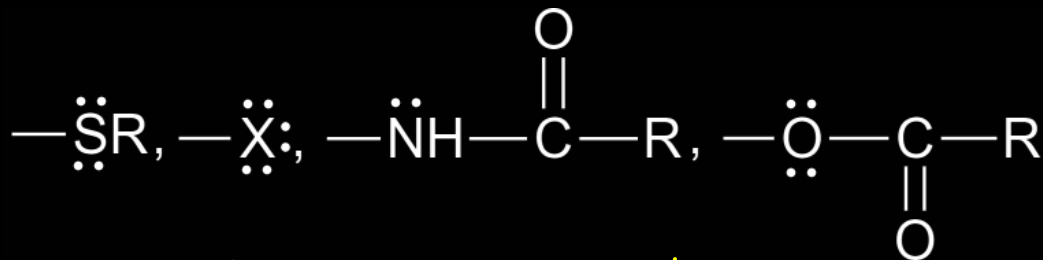
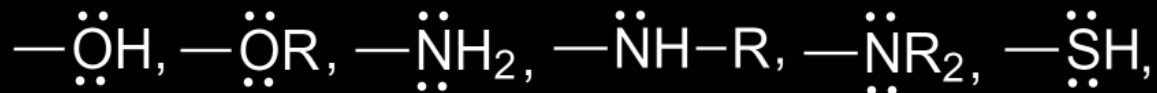
👉 +R OR +M effect :



An effect in which an atom or a group of atom donates e^- pair to the conjugated system through resonance is termed as +R / +M effect.

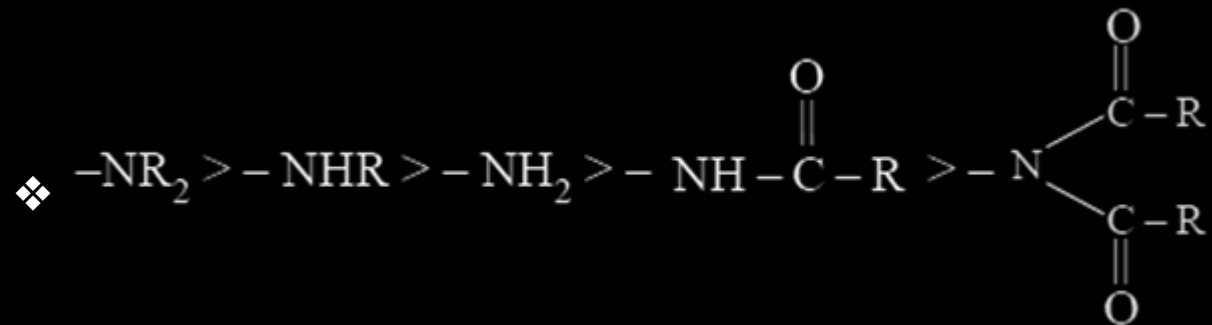
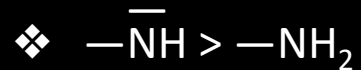
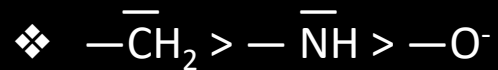
Mesomeric effect (R/M) :

Some more groups which shows +R effect when attached to conjugated system are :

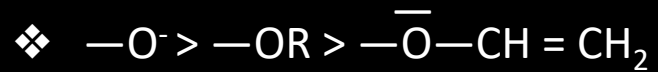


atom $\rightarrow$ lone pair

+ M effect order

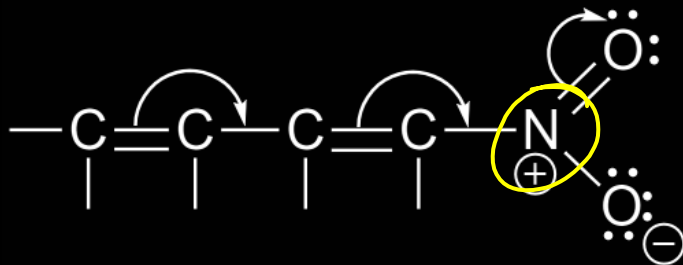


+ M effect order



Mesomeric effect (R/M) :

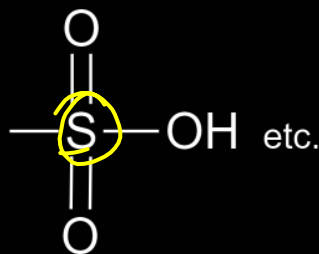
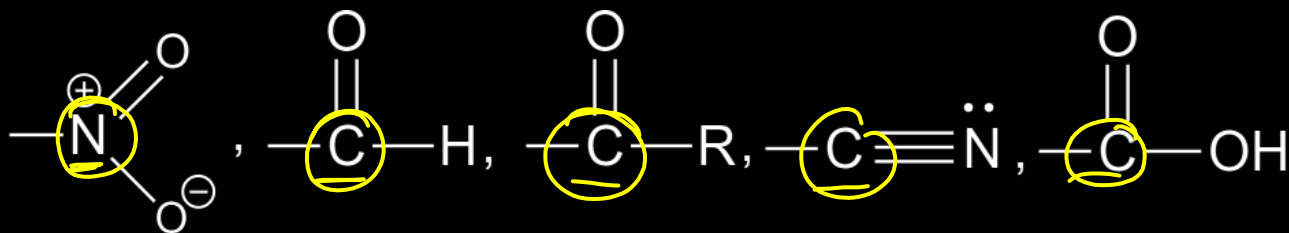
👉 -R OR -M effect :



An effect in which an atom or a group of atom attached to the conjugated system withdraws e⁻ pair through resonance is termed as -R / -M effect.

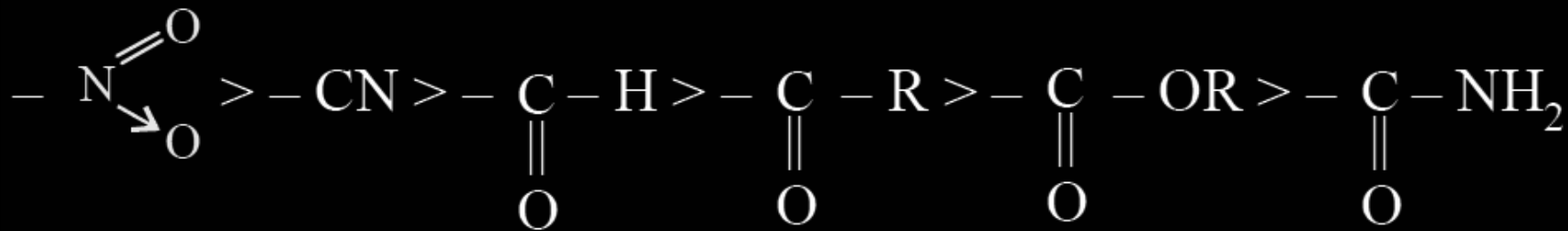
Mesomeric effect (R/M) :

Some more groups which shows –R attached to conjugated system are :



{ multiple bonded
electronegative atom

- M effect order



Mesomeric effect (R/M) :

Note :

- ❑ Thus, mesomeric effect works at only ortho & para position, it is absent on meta position, while inductive effect works at all three position o, m, p however intensity of effect decreases as the distance increases.
- ❑ Mesomeric effect always dominates on inductive effect except halogen (Cl, Br, I, only).

Resonance/
Mesomeric effect
(π effect)



Inductive
effect
(σ effect)

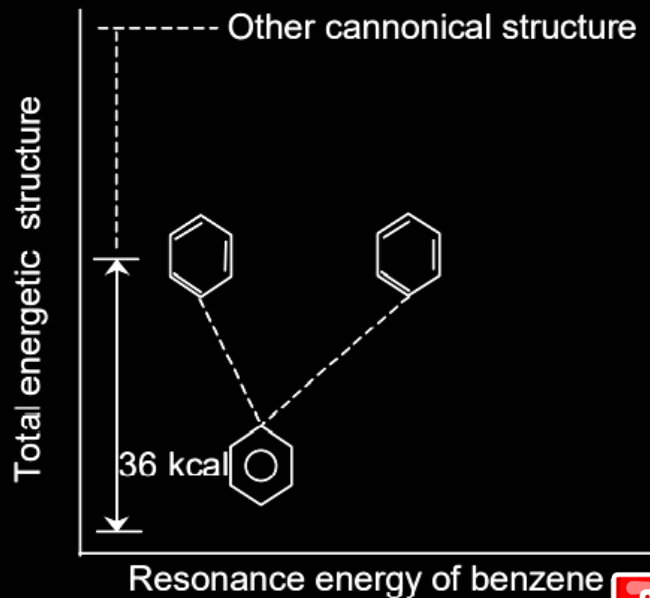
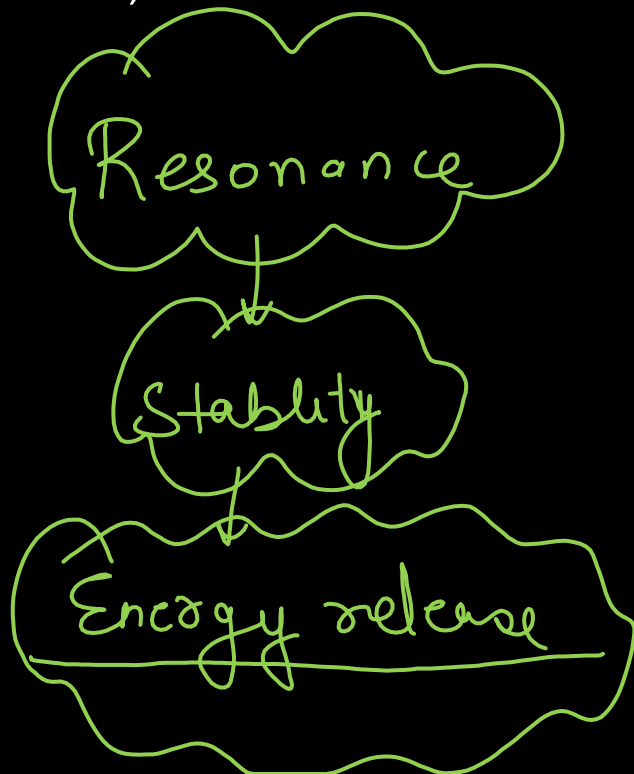
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Resonance Energy



Resonance Energy :

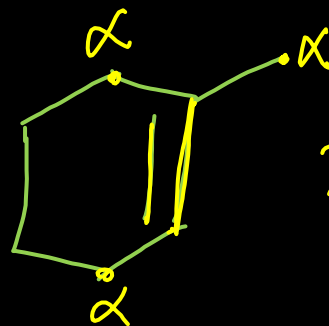
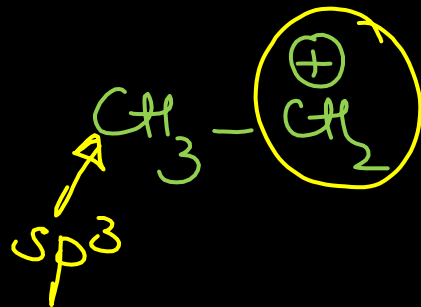
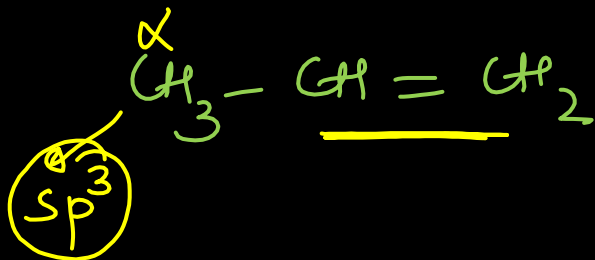
It is the difference in the energy between the resonance hybrid and most stable resonating structure further, more is the resonance energy more stable is the molecule.



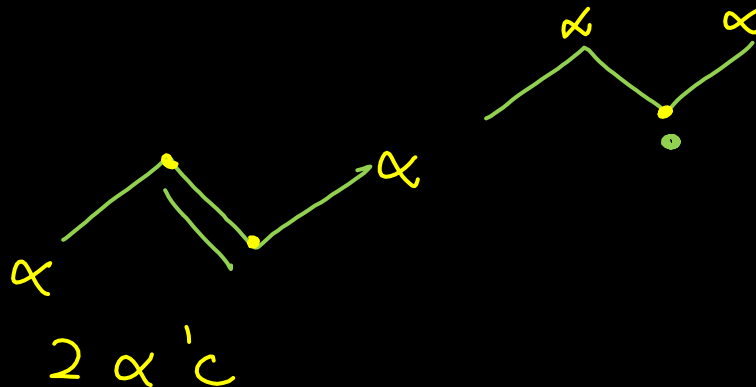
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Hyperconjugation

alpha C and H



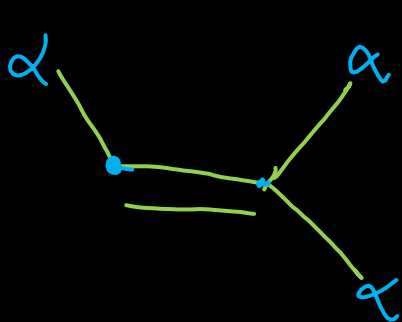
3 α C



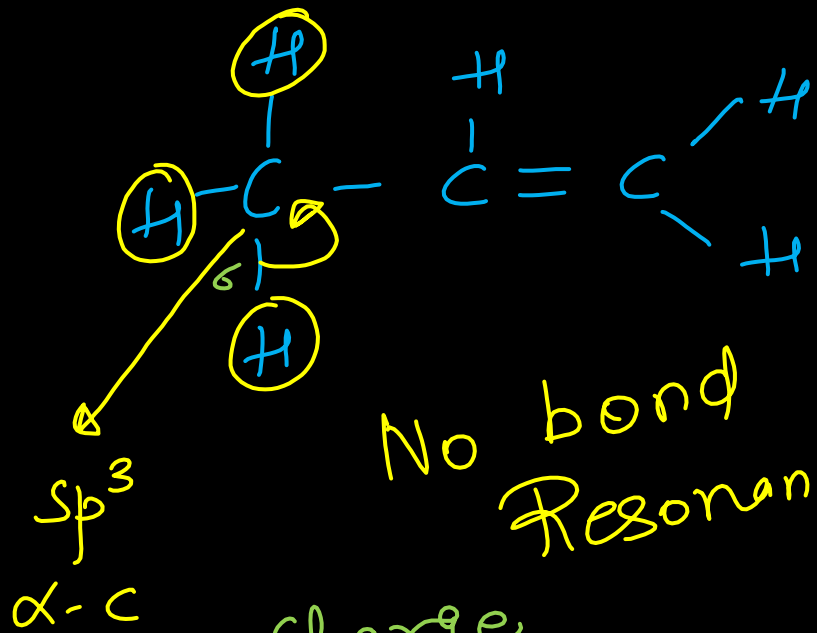
Hyperconjugation

Condition

→ presence of sp^3 hybridised α -C with +
double bond (alkene), carbocation, carbon
free radical



3 α C's
9 α H's

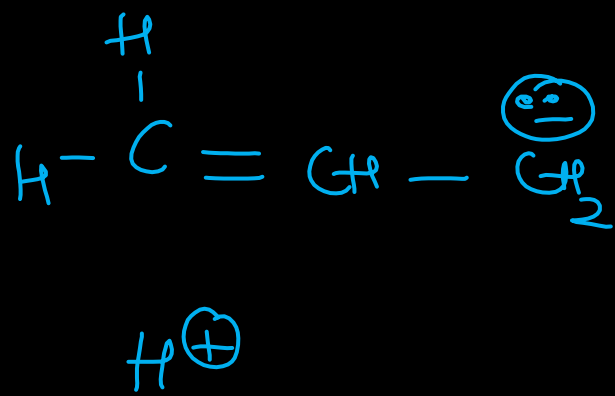
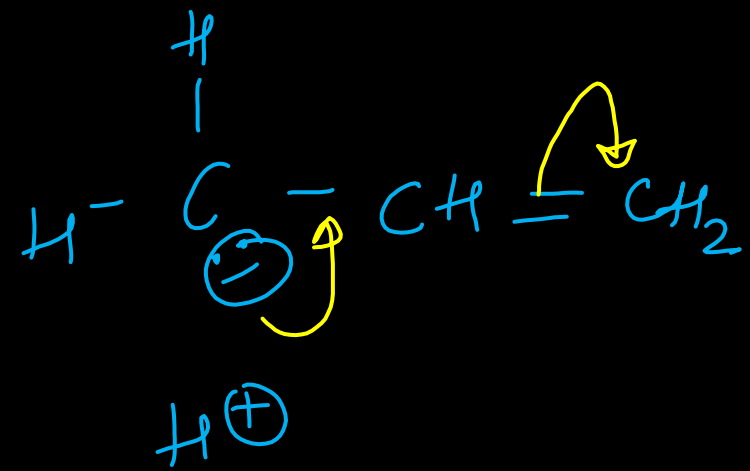


No bond
Resonance

charge
delocalisation

(σ, π -effect)

"Baker Nathan Effect"

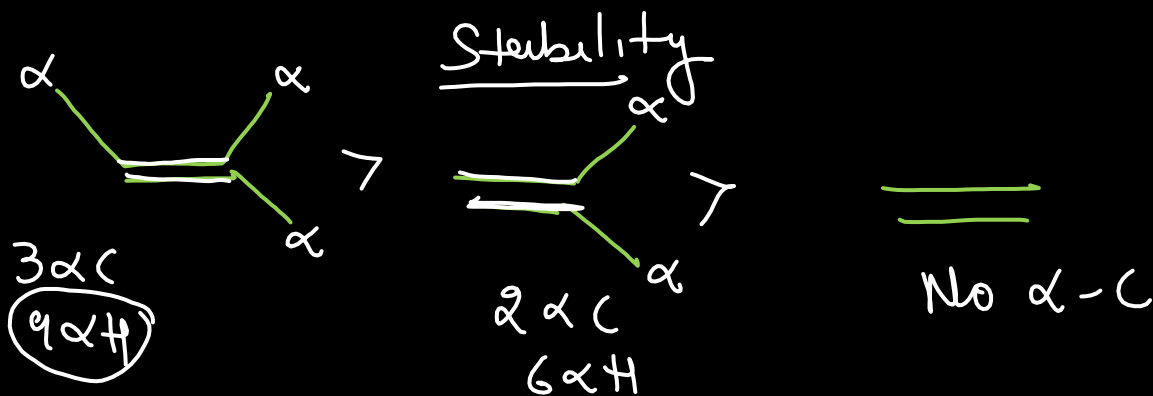


more α -H's $\Rightarrow$ more hyperconjugation



more stability

Remember



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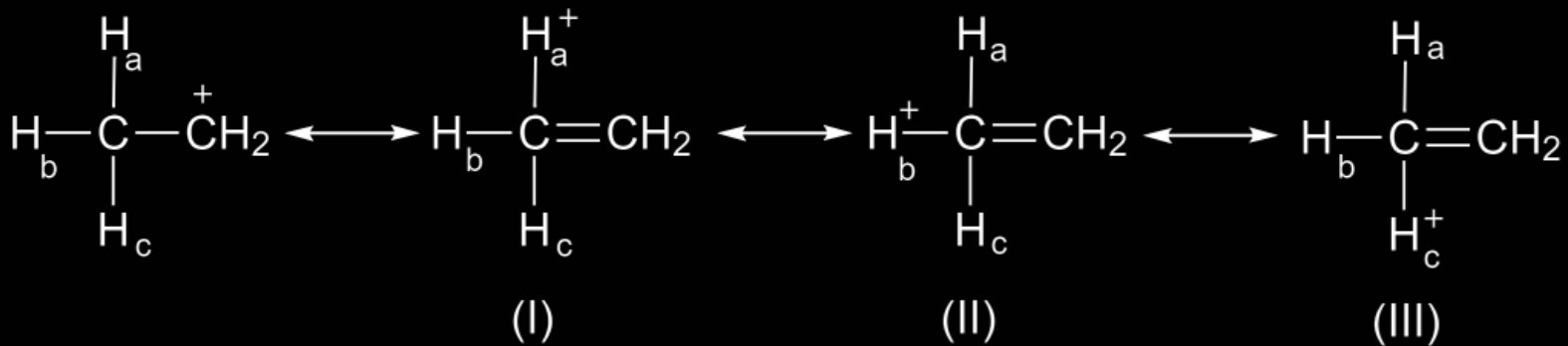
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BIGGEST EVENT
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Hyperconjugation

Representation / σ - π Conjugation /Nathan Baker Effect



APPLICATION OF HYPERCONJUGATION

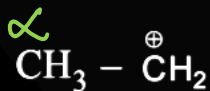
Stability of Alkyl Carbocations :

Stability of alkyl carbocations $\propto$ number of hyperconjugating structures number of hydrogens



(I)

no. of α H = 0



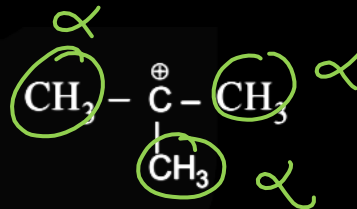
(II)

no. of α H = 3



(III)

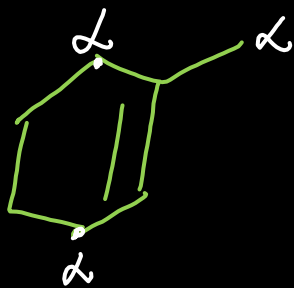
no. of α H = 6



(IV)

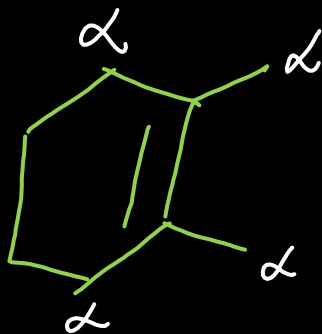
no. of α H = 9

stability order IV > III > II > I



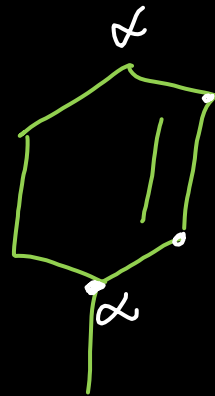
7 α H

(I)



10 α H's

(II)



3 α H

(III)

(I) > (II) >

stability

APPLICATION OF HYPERCONJUGATION

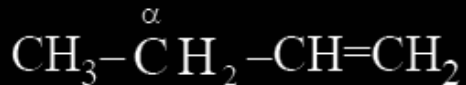
Stability of Alkenes :Hyperconjugation explains the stability of certain alkenes over other alkenes:

Stability of alkenes $\propto$ Number of alpha hydrogens $\propto$ Number of hyperconjugating structures

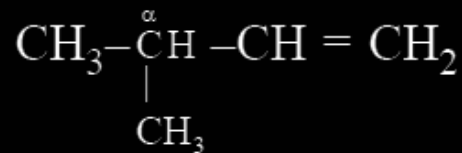
Ex.



(I)



(II)

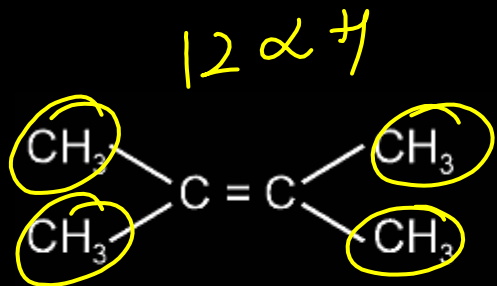


(III)

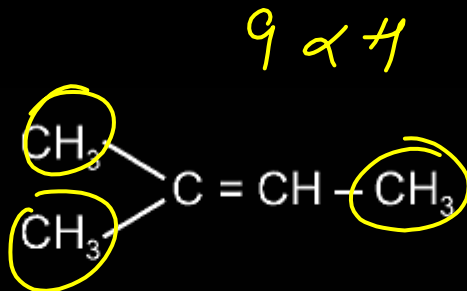
stability order $\text{I} > \text{II} > \text{III}$

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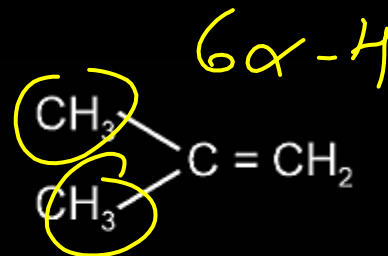
APPLICATION OF HYPERCONJUGATION



(I)



(II)



(III)

stability order $I > II > III$

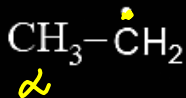
APPLICATION OF HYPERCONJUGATION

Stability of Alkyl free radicals : Stability of alkyl free radicals can be explained by hyperconjugation. Stability depends on the number of hyperconjugating structures.
structure



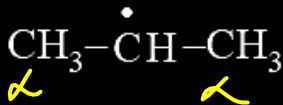
(I)

no. of α H = 0



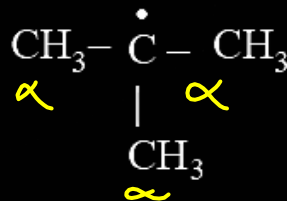
(II)

no. of α H = 3



(III)

no. of α H = 6



(IV)

no. of α H = 9

👉 Stability is in increasing order $\text{IV} > \text{III} > \text{II} > \text{I}$

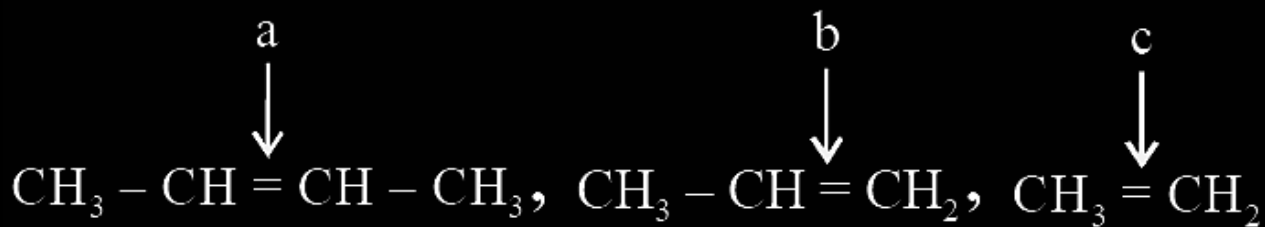
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APPLICATION OF HYPERCONJUGATION

Bond length in alkenes: More is the number of hyperconjugating structures, the more will be single bond character in carbon-carbon double bond.

The bond length between carbon-carbon double bond $\propto$ number of hyperconjugating structures.

Ex.

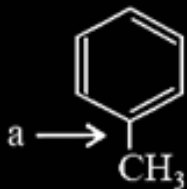


Bond length order $\Rightarrow a > b > c$

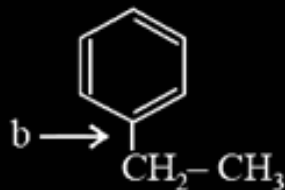


APPLICATION OF HYPERCONJUGATION

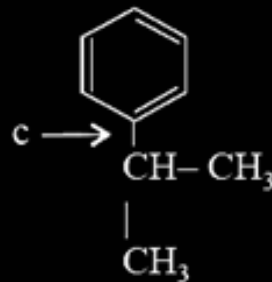
Ex.



,



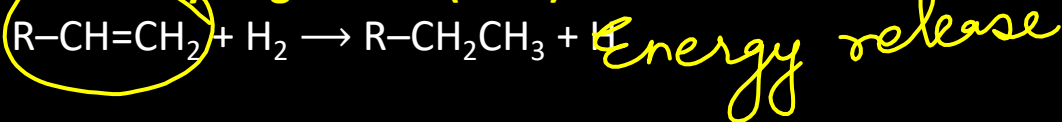
,



Bond length order $\Rightarrow a > b > c$

APPLICATION OF HYPERCONJUGATION

Heat of hydrogenation (HOH) :

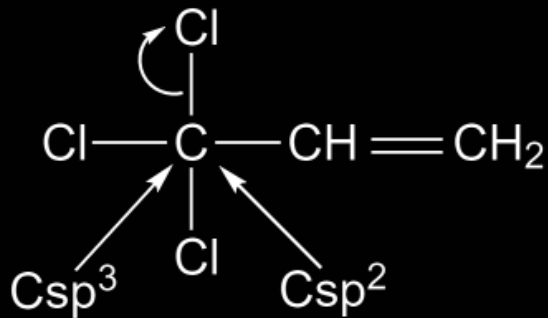


- ❖ heat evolved when 1 mole of any unsaturated hydrocarbon are hydrogenated is called heat of hydrogenation (ΔH) If alkene is more reactive towards hydrogen then it will evolve more ΔH .

So,
$$\Delta H \propto \frac{1}{\text{stability of alkene}} \propto \frac{1}{\text{number of } \alpha\text{-H}}$$

heat of hydrogenation

Reverse Hyper conjugation



π bond of C_{sp^2} Delocalized toward C_{sp^3} C atom.

–Reverse Hyper conjugation (–H)

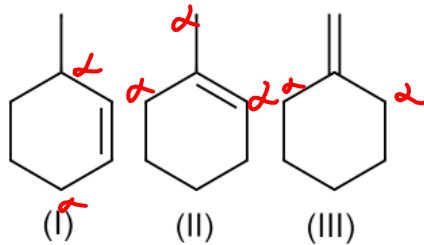
Stability of alkenes and Heat of hydrogenation :

$$\text{Heat of hydrogenation} \propto \frac{1}{\text{Stability of alkenes}}$$

Remember

GOC → strong

Find the order of stability of following alkenes:



A

II > III > I

B

I > II > III

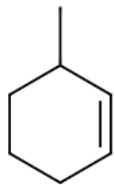
C

III > II > I

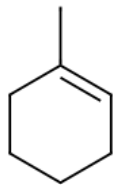
D

I > III > II

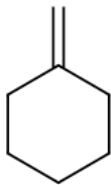
Find the order of stability of following alkenes:



(I)



(II)



(III)

A

II > III > I

B

I > II > III

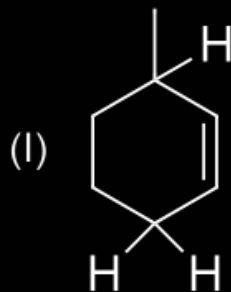
C

III > II > I

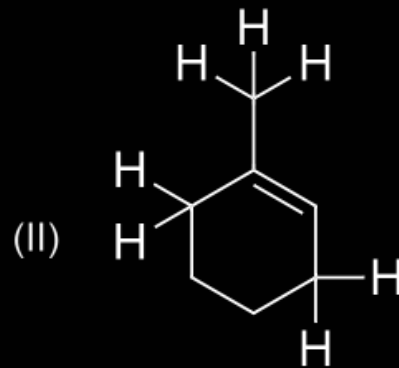
D

I > III > II

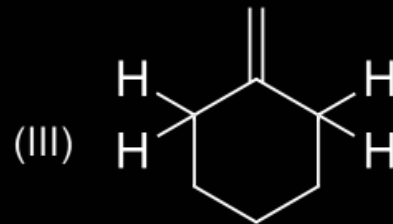
Solution :



$3\alpha\text{-H'S}$



$7\alpha\text{-H'S}$

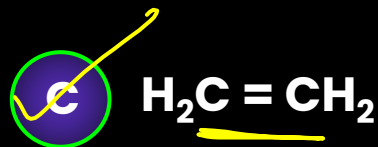
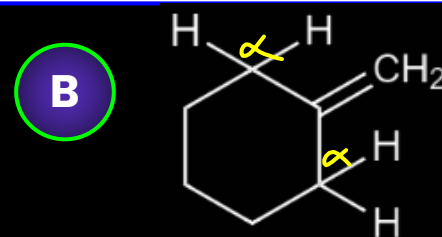
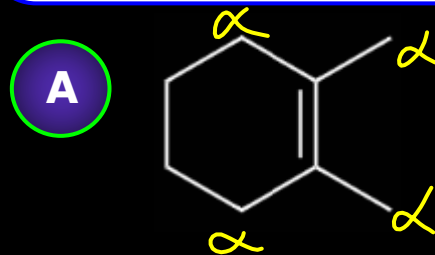


$4\alpha\text{-H'S}$

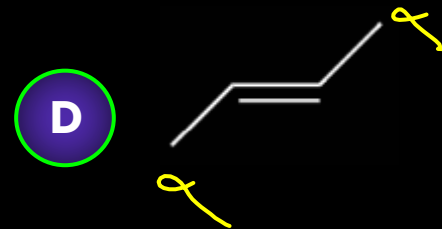
Thus the order of stability = II > III > I

In which of the following alkenes hyperconjugation is not possible?

yf no α -C

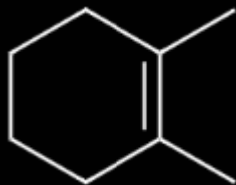


no α -H

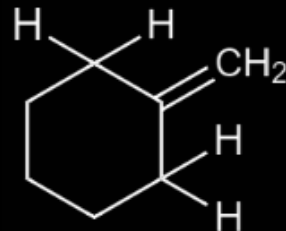


In which of the following alkenes hyperconjugation is not possible?

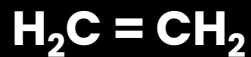
A



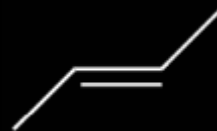
B



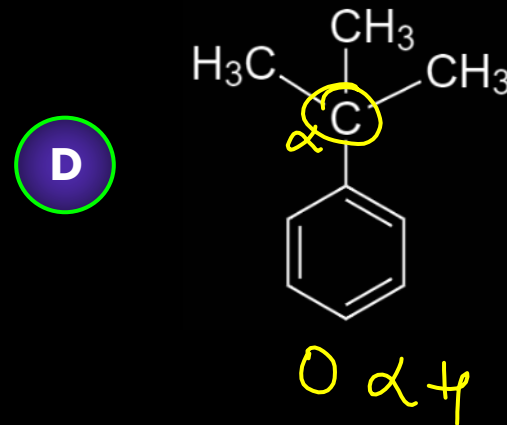
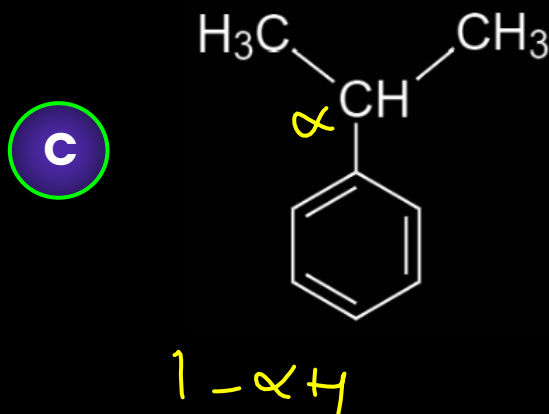
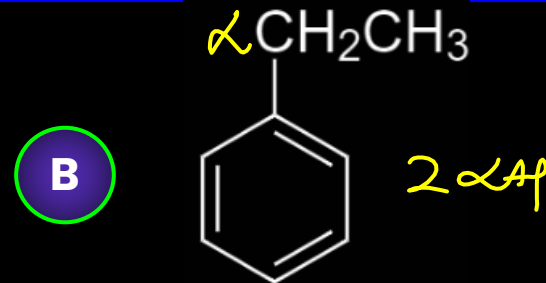
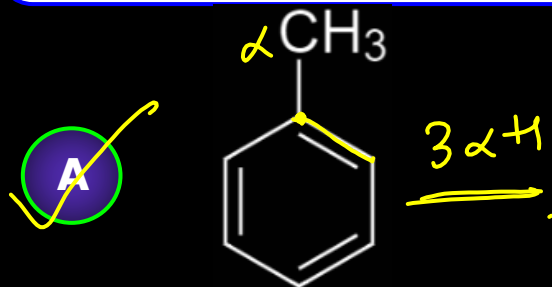
C



D

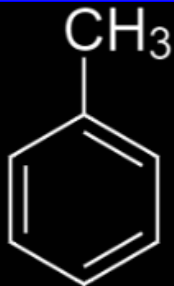


Which of the following is most stabilised by hyperconjugation?

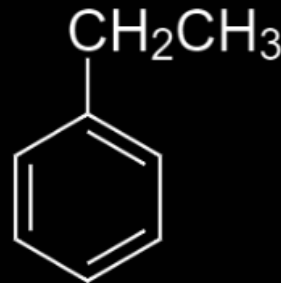


Which of the following is most stabilised by hyperconjugation?

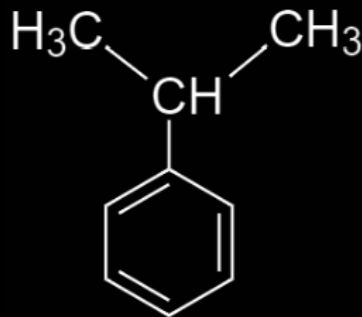
A



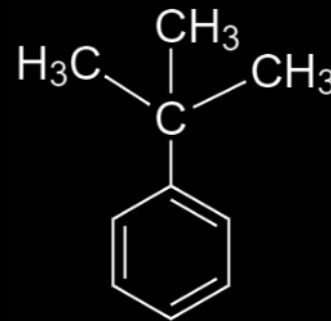
B



C



D



Impact of Electronic Effects of Substituents on the Stability of Conjugate Base/Acid



Resonance > Hyperconjugation > Inductive

Priority Order

^{Resonance}
M effect > I-effect

(except Halogens)

(I) > (M effect)

Note

For a acid / Basic
Strength

Acidic Strength

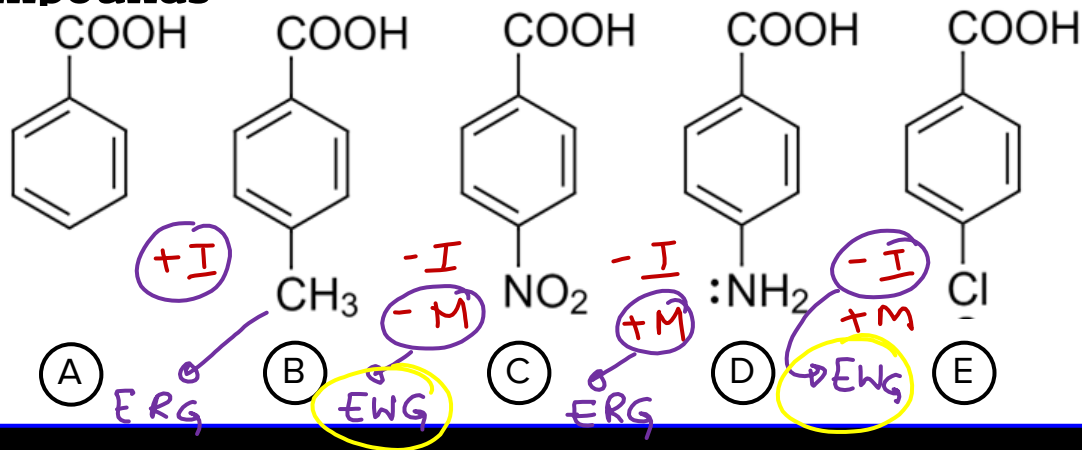
$$\text{Acidic strength} \propto \text{stability of conjugate base} \propto -M \text{ effect} \propto \frac{I}{+M} \propto -I \propto \frac{1}{+I}$$

→ $M \text{ effect} > I \text{ effect}$ except halogens

→ $\boxed{\text{EWG}(-M|-I)} \Rightarrow \underline{\text{increases acidic strength}}$

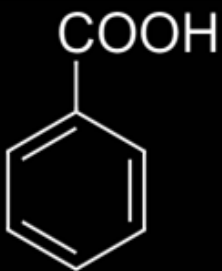
$\boxed{\text{ERG}(+M|+I)} \Rightarrow \underline{\text{decreases acidic strength}}$

Write the order of acidic strength of the following compounds

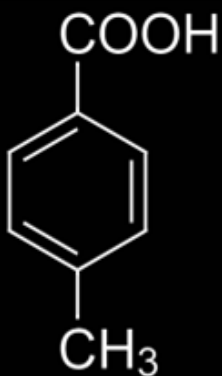


Acidic strength

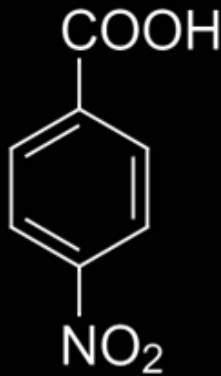
Solution :



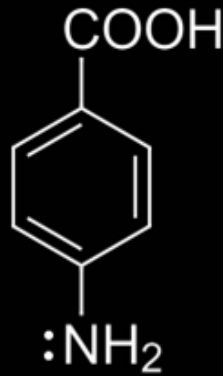
A



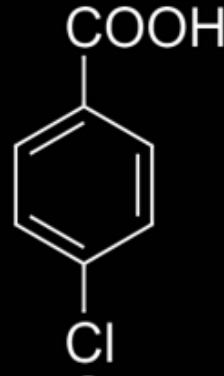
B



C



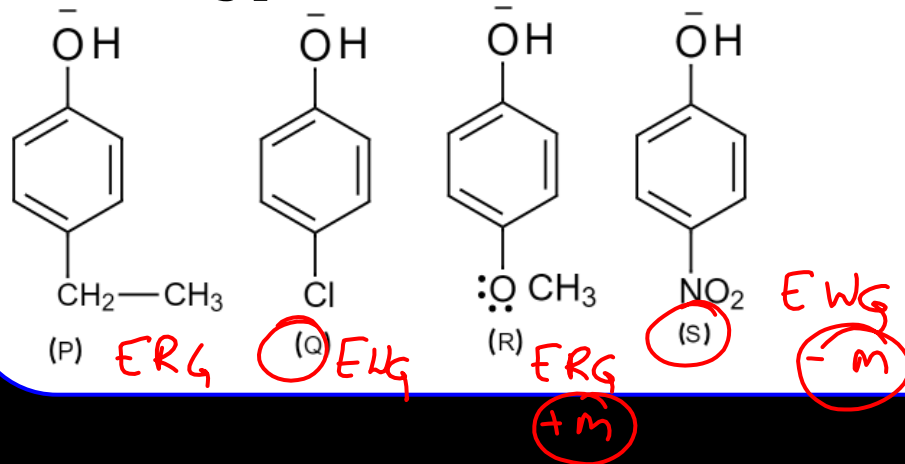
D



E

Compound is acidic if it's conjugate base is stable, further more stable is the conjugate base stronger is the acid.

Write the decreasing order of acidic strength of following phenol derivative :



~~A~~

$\text{S} > \text{Q} > \text{P} > \text{R}$

B

$\text{P} > \text{Q} > \text{R} > \text{S}$

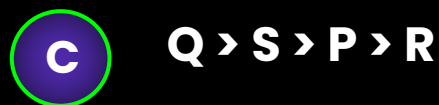
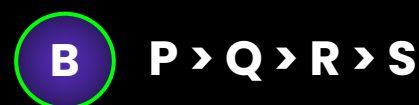
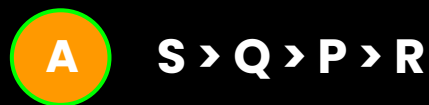
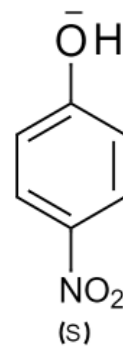
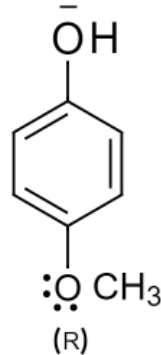
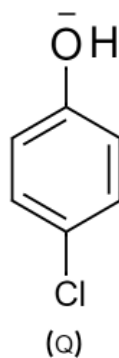
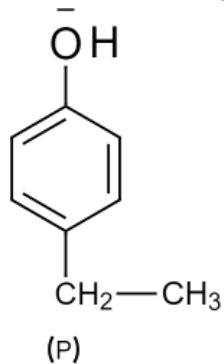
C

$\text{Q} > \text{S} > \text{P} > \text{R}$

D

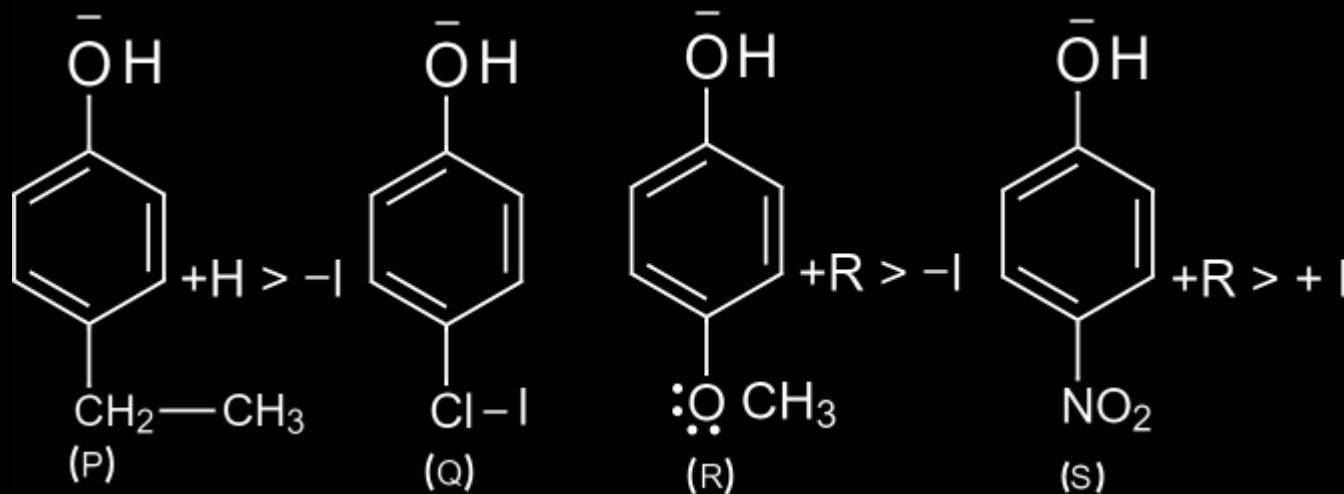
$\text{S} > \text{R} > \text{Q} > \text{P}$

Write the decreasing order of acidic strength of following phenol derivative :



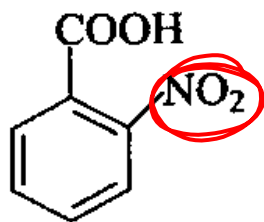
Write the decreasing order of acidic strength of following phenol derivative :

Solution :

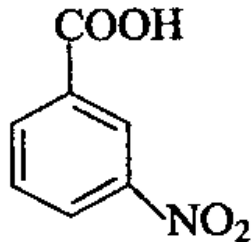


Thus, the order is : $S > Q > P > R$

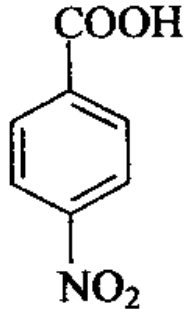
The decreasing order acidity of the given benzoic acid derivative is :



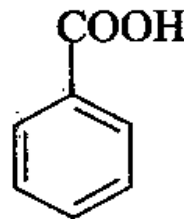
(P)



(Q)



(R)



(S)

A

P > Q > R > S

B

P > R > Q > S

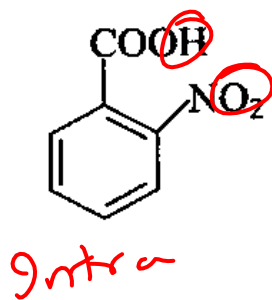
C

R > P > Q > S

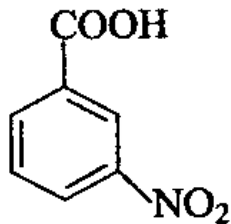
D

S > R > Q > P

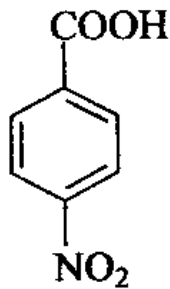
The decreasing order acidity of the given benzoic acid derivative is :



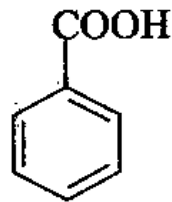
(P)



(Q)



(R)



(S)

A

P > Q > R > S

B

P > R > Q > S

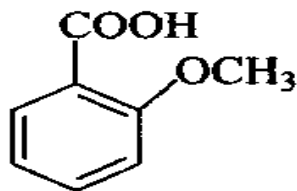
C

R > P > Q > S

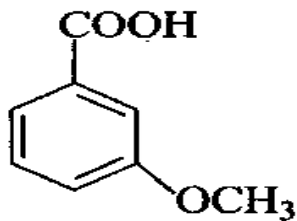
D

S > R > Q > P

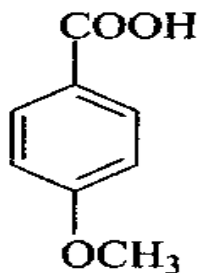
The decreasing order acidity of the given benzoic acid derivative is :



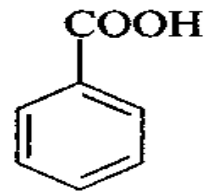
(P)



(Q)



(R)



(S)

A

S > Q > R > P

B

R > Q > P > S

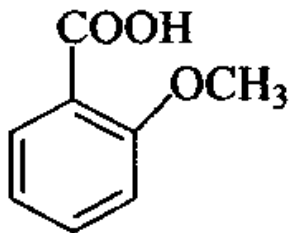
C

Q > R > S > P

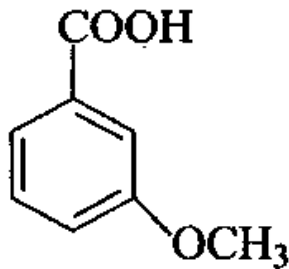
D

P > Q > S > R

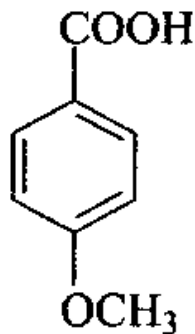
The decreasing order acidity of the given benzoic acid derivative is :



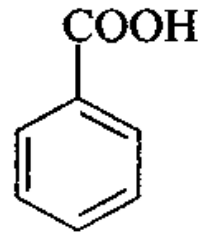
(P)



(Q)



(R)



(S)

A

S > Q > R > P

B

R > Q > P > S

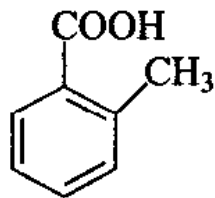
C

Q > R > S > P

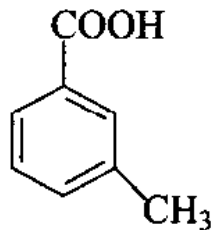
D

P > Q > S > R

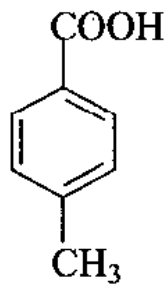
The decreasing order acidity of the given benzoic acid derivative is :



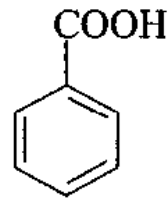
(P)



(Q)



(R)



(S)

A

P > R > Q > S

B

S > R > Q > P

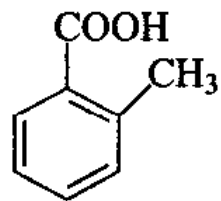
C

P > S > Q > R

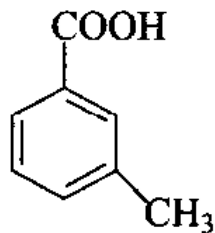
D

Q > S > P > R

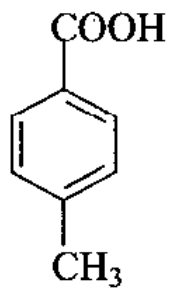
The decreasing order acidity of the given benzoic acid derivative is :



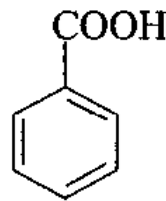
(P) ✓



(Q)



(R)



(S)

A

P > R > Q > S

B

S > R > Q > P

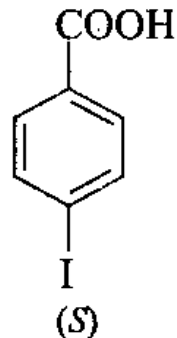
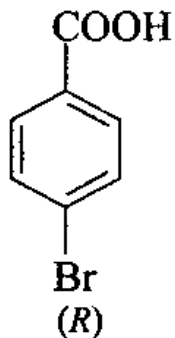
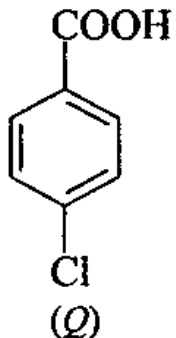
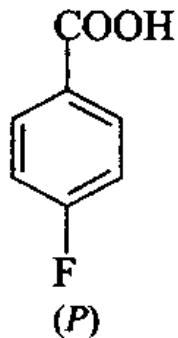
C

P > S > Q > R

D

Q > S > P > R

The decreasing order acidity of the given benzoic acid derivative is :



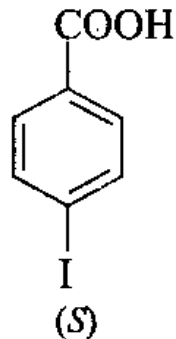
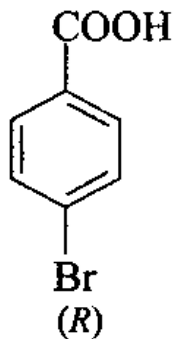
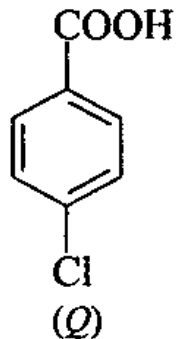
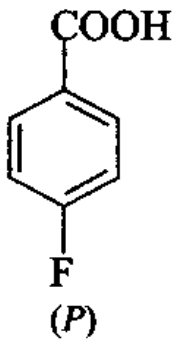
A P > Q > R > S

B S > R > Q > P

C R > Q > P > R

D Q > P > S > R

The decreasing order acidity of the given benzoic acid derivative is :



A

P > Q > R > S

B

S > R > Q > P

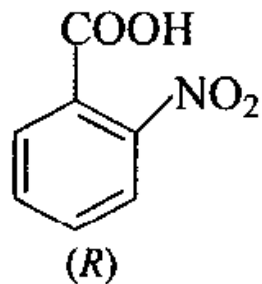
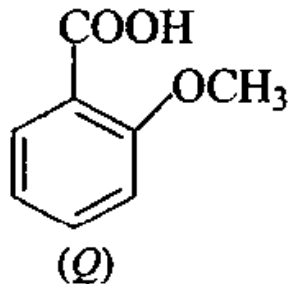
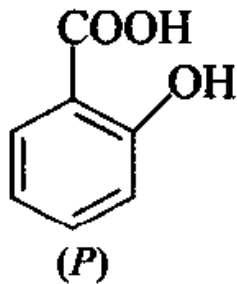
C

R > Q > P > S

D

Q > P > S > R

The decreasing order acidity of the given benzoic acid derivative is :



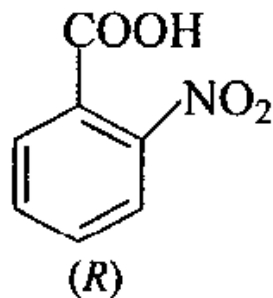
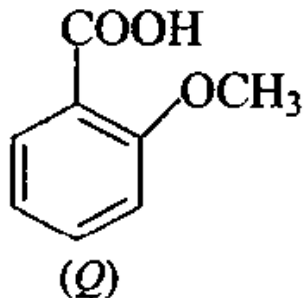
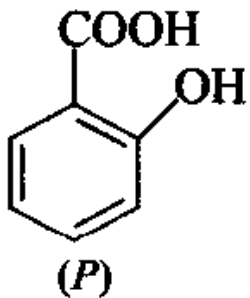
A $P > Q > R$

B $R > P > Q$

C $Q > R > P$

D $P > R > Q$

The decreasing order acidity of the given benzoic acid derivative is :



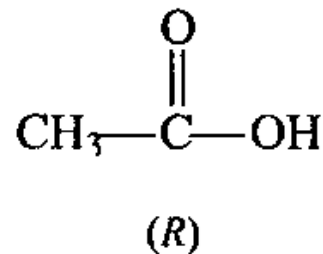
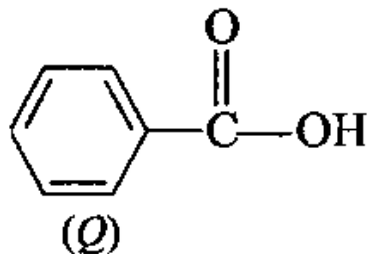
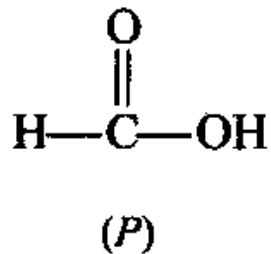
A $P > Q > R$

B $R > P > Q$

C $Q > R > P$

D $P > R > Q$

The decreasing order acidity of the given benzoic acid derivative is :



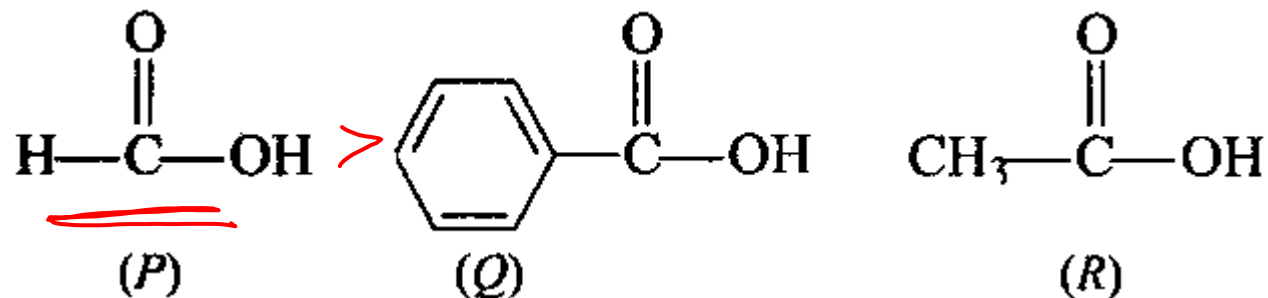
A $R > Q > P$

B $Q > P > R$

C $P > Q > R$

D $Q > R > P$

The decreasing order acidity of the given benzoic acid derivative is :



Keep in mind

A R > Q > P

B Q > P > R

C P > Q > R

D Q > R > P

Basic Strength

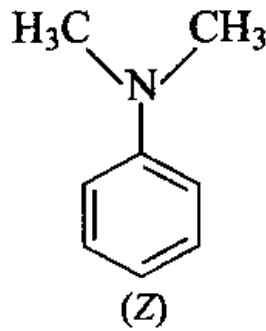
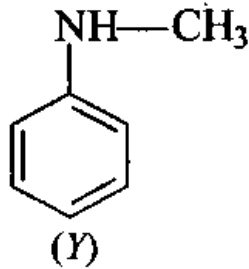
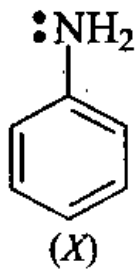
Base $\rightarrow$ e^- donate

$$\text{basic strength} \propto +M \propto \frac{1}{-M} \propto +I \propto \frac{1}{-I}$$

EWG ($-M/-I$) $\Rightarrow$ decreases basic strength

ERG ($+M/+I$) $\Rightarrow$ increases basic strength

The decreasing order of basicity of the given aniline derivative is :



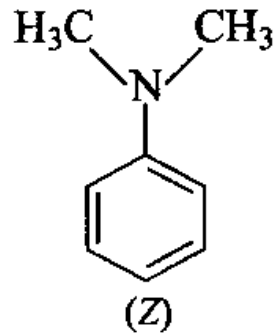
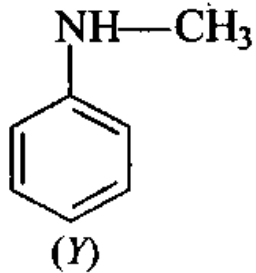
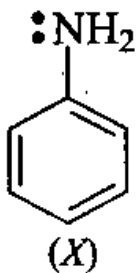
A $X > Y > Z$

B $Y > Z > X$

C $Z > Y > X$

D $X > Z > Y$

The decreasing order of basicity of the given aniline derivative is :



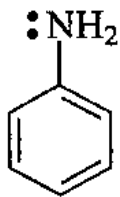
A $X > Y > Z$

B $Y > Z > X$

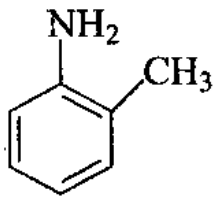
C $Z > Y > X$

D $X > Z > Y$

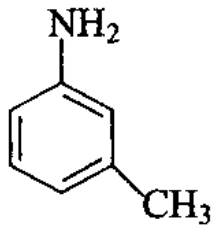
The decreasing order of basicity of the given aniline derivative is :



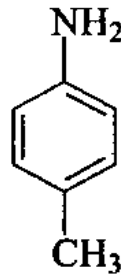
(P)



(Q)



(R)



(S)

A

S > R > P > Q

B

S > R > Q > P

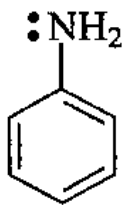
C

P > Q > R > S

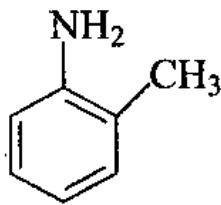
D

Q > S > R > P

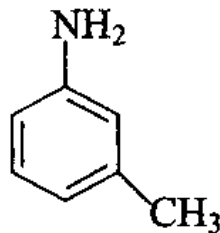
The decreasing order of basicity of the given aniline derivative is :



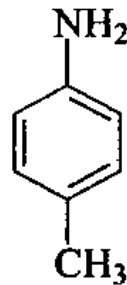
(P)



(Q)



(R)



(S)

A

$\text{S} > \text{R} > \text{P} > \text{Q}$

B

$\text{S} > \text{R} > \text{Q} > \text{P}$

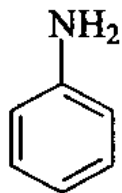
C

$\text{P} > \text{Q} > \text{R} > \text{S}$

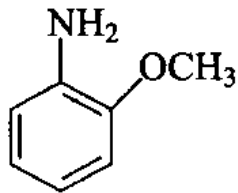
D

$\text{Q} > \text{S} > \text{R} > \text{P}$

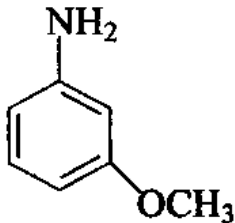
The decreasing order of basicity of the given aniline derivative is :



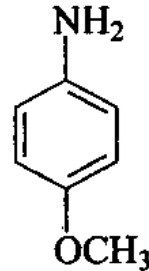
(P)



(Q)



(R)



(S)

A

S > R > Q > P

B

P > Q > R > S

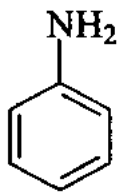
C

S > Q > P > R

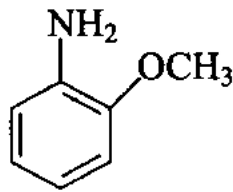
D

S > P > Q > R

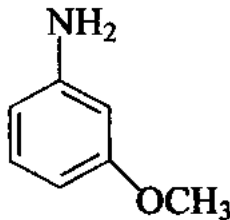
The decreasing order of basicity of the given aniline derivative is :



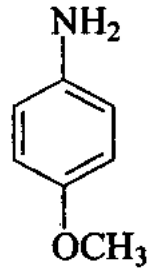
(P)



(Q)



(R)



(S)

A

S > R > Q > P

B

P > Q > R > S

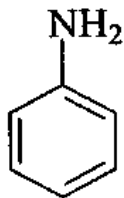
C

S > Q > P > R

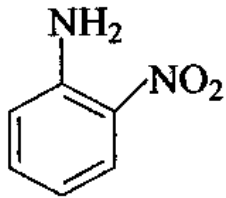
D

S > P > Q > R

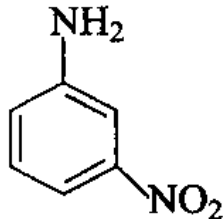
The decreasing order of basicity of the given aniline derivative is :



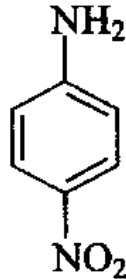
(P)



(Q)



(R)



(S)

A

P > Q > R > S

B

P > R > S > Q

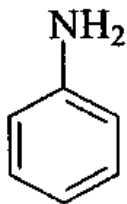
C

R > Q > P > S

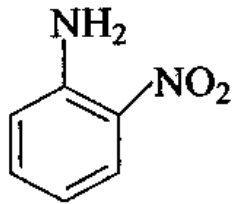
D

S > R > Q > P

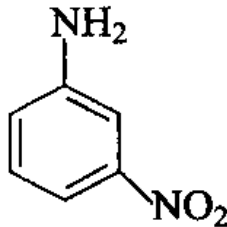
The decreasing order of basicity of the given aniline derivative is :



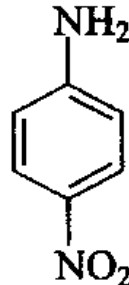
(P)



(Q)



(R)



(S)

A

$\text{P} > \text{Q} > \text{R} > \text{S}$

B

$\text{P} > \text{R} > \text{S} > \text{Q}$

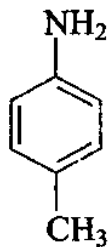
C

$\text{R} > \text{Q} > \text{P} > \text{S}$

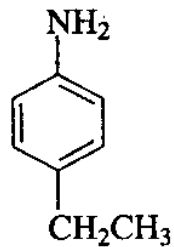
D

$\text{S} > \text{R} > \text{Q} > \text{P}$

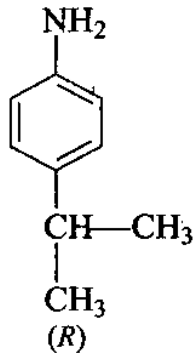
The decreasing order of basicity of the given aniline derivative is :



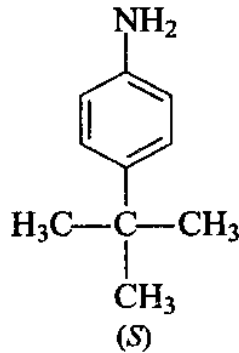
(P)



(Q)



(R)



(S)

A

P > Q > R > S

B

Q > S > R > P

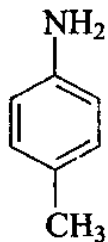
C

S > R > Q > P

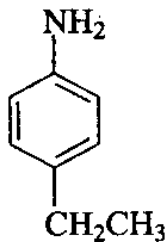
D

R > S > Q > P

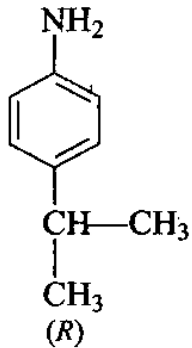
The decreasing order of basicity of the given aniline derivative is :



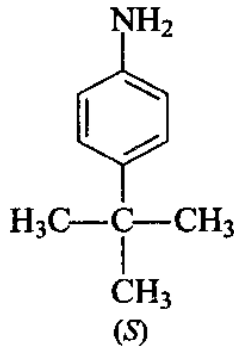
(P)



(Q)



(R)



(S)

A

P > Q > R > S

B

Q > S > R > P

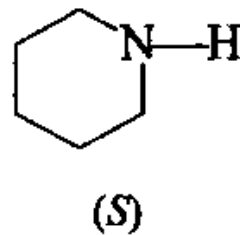
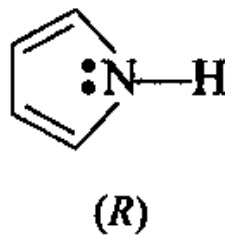
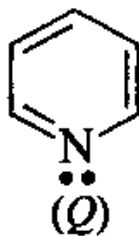
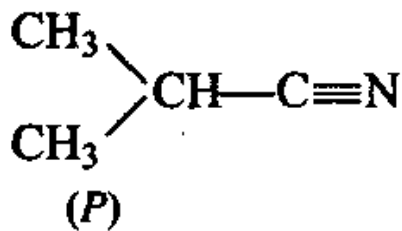
C

S > R > Q > P

D

R > S > Q > P

The decreasing order of basicity of the given aniline derivative is :



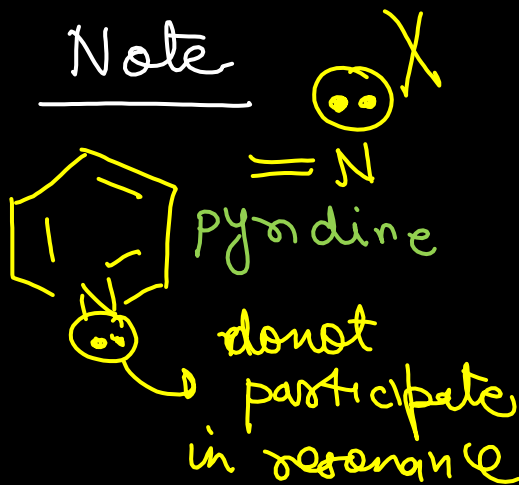
A R > Q > S > P

B P > Q > R > S

C Q > P > S > R

D S > Q > R > P

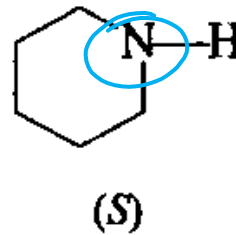
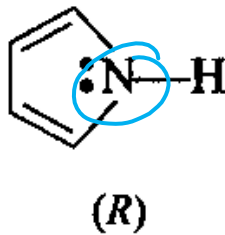
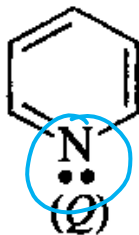
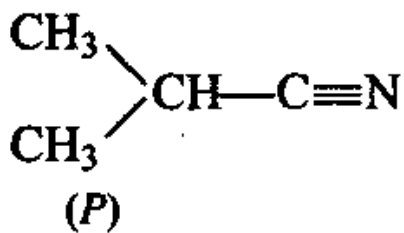
Note



pyridine

do not participate in resonance

The decreasing order of basicity of the given aniline derivative is :



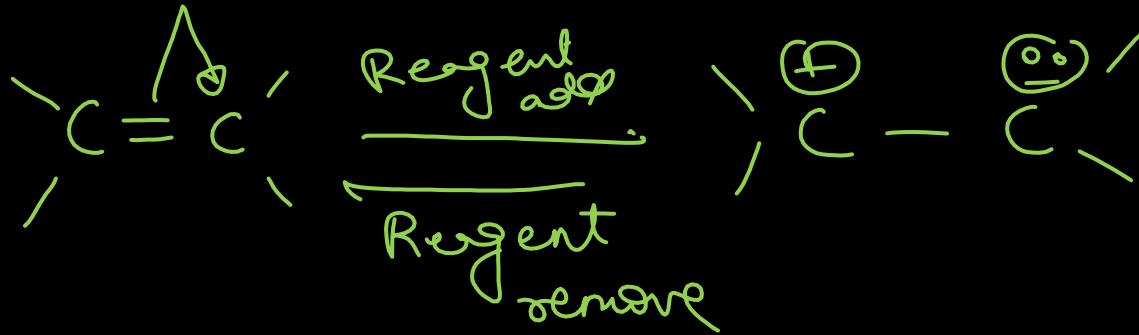
A R > Q > S > P

B P > Q > R > S

C Q > P > S > R

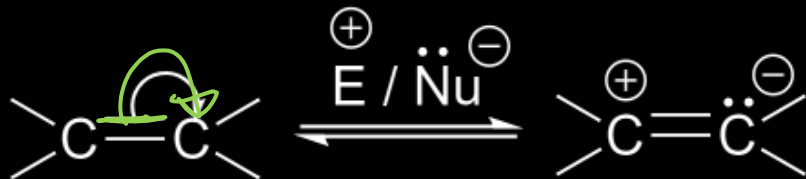
D S > Q > R > P

Electromeric Effect



Electromeric Effect :

Representation :



Temporary complete transfer of πe^- s in presence of attacking reagent.

Temporary complete transfer of πe^- s towards preferentially more EN atom in presence of attacking reagent.

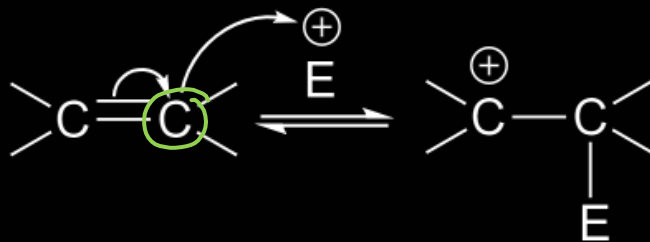
Electromeric Effect :

Characteristics :

- 👉 It's a temporary effect.
- 👉 It's a complete transfer of πe^- s.
- 👉 Presence of attacking reagent is must for this effect to takes place.

Types of Electromeric Effects (+E)

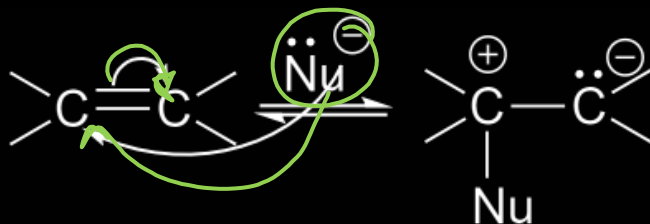
+ E effect :



When the transfer of e^- s takes place at the atom of attacking reagent, it is termed as "Electromeric Effect".

Types of Electromeric Effects (+E)

– E effect :



When the transfer of e^- s takes place away from the atom of attacking reagent, it is termed as $-E$ Effect.

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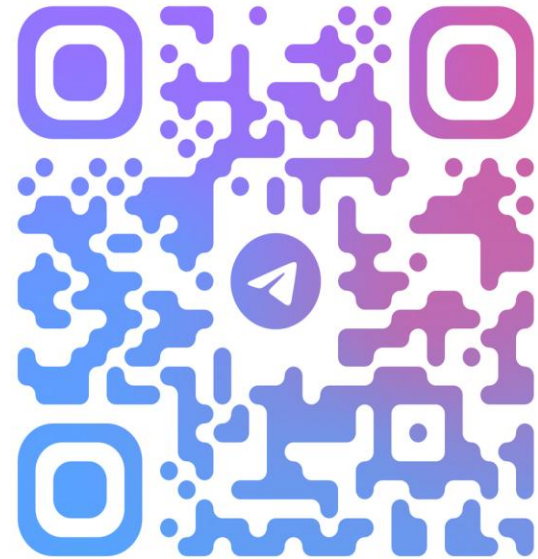


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